# HIKARI: A Multi-Stage Vision-Language Prototype with RAG-in-Training for Medical Lesion Diagnosis

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***Abstract*—We present HIKARI (Healthcare-oriented Integrated Knowledge-Augmented Retrieval and Inference system), a prototype multi-stage Vision-Language Model (VLM) pipeline for medical lesion diagnosis and clinical caption generation. Retrieval-Augmented Generation (RAG) provides visual few-shot context at inference time, yet a train-inference distribution gap persists when the model is fine-tuned on single images. We propose RAG-in-Training, which incorporates retrieved reference images into every training sample, eliminating this gap. Built on a three-stage cascaded Low-Rank Adaptation (LoRA) fine-tuning pipeline using Qwen3-VL-8B-Thinking on the SkinCAP dataset (10 disease classes, 99 validation samples), HIKARI achieves 85.86% classification accuracy (95% Wilson CI [77.7–91.4%]), surpassing the strongest CNN baseline (YOLOv11, 65.00%; CI [55.2–73.7%]) by 20.86 pp, single-image FT by 11.86 pp, and inference-only RAG by 6.06 pp (directional; CI includes 0). Merging adapter weights before task switching (Merged-Init) yields a 3× BLEU-4 gain (9.82→29.33), ROUGE-1: 38.9→53.6 (+14.7 pp), and BERTScore $F_1$ 92.1% [24].**

***Index Terms*—medical lesion classification, retrieval-augmented generation, vision-language model, LoRA fine-tuning, multi-stage training, clinical caption generation, clinical dermatology, HIKARI**

## I. Introduction

Medical lesion diagnosis from clinical images remains challenging for fine-grained inflammatory diseases (psoriasis, lupus, lichen planus) with limited per-class data and subtle inter-class differences. Analogous challenges arise in oral mucosal disease [10], diabetic retinopathy, and colonoscopy grading.

We present **HIKARI**, a prototype multi-stage vision-language pipeline for (1) fine-grained lesion classification, (2) clinical caption generation, and (3) closing the train-inference distribution gap when RAG is applied only at inference. This gap is especially harmful in medical imaging: a model that has never seen reference-augmented inputs during training treats the retrieved image as visual noise rather than as diagnostic context. Validated on clinical skin lesion data, the pipeline is architecturally domain-agnostic—swapping SkinCAP's RAG index re-targets the disease domain without modifying the training procedure—though empirical cross-domain validation remains future work and is required before any generalization claim. Seven frontier VLMs without fine-tuning score only 12.87–50.51%; even retrieval-augmented Qwen models reach 33.33–50.51%, confirming that reference injection requires RAG-aware training [4].

HIKARI contributes: (1) **RAG-in-Training**—retrieved references in every training sample, closing the distribution gap from single-image FT (74.00%→85.86%; +11.86 pp) and surpassing the strongest CNN baseline (YOLOv11, 65.00%) by 20.86 pp; (2) **Hybrid RAG encoder evaluation** over five combinations, identifying image-dominant weighting ($\alpha$=0.9) for clinical skin lesion images; (3) **Merged-Init**—merging prior-stage adapters before task switching, yielding 3× BLEU-4 (9.82→29.33).

## II. Related Work

**Medical lesion classification.** Esteva et al. [2] demonstrated CNN-level dermatologist performance; transformer and contrastive pretraining approaches have since improved ISIC accuracy [1], yet VLM approaches beyond zero-shot remain underexplored for fine-grained clinical dermatology.

**RAG for medical imaging.** Lewis et al. [4] introduced RAG for knowledge-intensive NLP; multi-modal extensions

show gains in medical VQA [7]. Retrieval at inference without corresponding training-time exposure creates a distribution gap our work directly targets.

**Multi-stage LoRA fine-tuning.** Hu et al. [3] introduced LoRA for efficient VLM adaptation; cascaded application can cause catastrophic interference. OralGPT [10] shows sequential task specialization outperforms joint training in low-data domains. Our Merged-Init strategy absorbs prior-stage adapters into full-rank base weights before each new task, eliminating inter-task gradient interference.

**Efficient VLM serving.** PagedAttention [12] and structured generation scheduling [13] reduce large-model inference latency by 6–12× through memory-efficient KV-cache management and continuous batching. We evaluate both frameworks on HIKARI's two inference stages (classification and captioning), demonstrating that multi-stage clinical inference is practical on commodity hardware.

## III. Methodology

### A. Three-Stage Cascaded Fine-Tuning

Our pipeline progressively specializes a Qwen3-VL-8B-Thinking backbone [11] across three stages (Fig. 1).

**Stage 1** classifies into four coarse groups (88.68% coarse-group accuracy on the 99-sample validation set) with LoRA hyperparameters $r$=16, $s$=16, lr $2\times10^{-4}$, 3 epochs, batch 2, gradient accumulation 4 steps (AdamW-8bit); merged weights initialize Stage 2 (+4.6 pp over base initialization; Table VI). **Stage 2** fine-tunes for 10-class classification with LoRA hyperparameters: $r$=16, $s$=32, lr $2\times10^{-4}$, 3 epochs, AdamW-8bit, batch 8 on RTX 5070 Ti (15.92 GB VRAM).

**Stage 3** fine-tunes for clinical caption generation from the merged Stage 2 model (Section III-C) with LoRA hyperparameters $r$=16, $s$=16, lr $2\times10^{-4}$, 1 epoch, batch 2, gradient accumulation 4 steps (AdamW-8bit).

The **Single-Image FT** baseline fine-tunes Qwen3-VL-8B-Thinking directly on the 10-class Stage 2 task without group pre-training (Stage 1) or retrieved references, serving as an ablation for both components. Table I summarises all per-stage hyperparameters.

TABLE I
PER-STAGE LoRA HYPERPARAMETERS. GRAD. ACC. = GRADIENT ACCUMULATION STEPS. ONLY STAGE 2 USES RAG-IN-TRAINING ($K$=1).

| | **Stage 1** | **Stage 2** | **Stage 3** |
|---|---|---|---|
| Task | 4-group class. | 10-class class. | Caption gen. |
| LoRA $r$ | 16 | 16 | 16 |
| LoRA $s$ | 16 | 32 | 16 |
| Learning rate | $2\times10^{-4}$ | $2\times10^{-4}$ | $2\times10^{-4}$ |
| Epochs | 3 | 3 | 1 |
| Batch size | 2 | 8 | 2 |
| Grad. acc. | 4 | 1 | 4 |
| RAG $K$ (train) | – | 1 | – |
| Init | Base | Merged-S1 | Merged-S2 |

### B. RAG-in-Training

**Hybrid RAG index.** For each training image we pre-compute embeddings using a hybrid encoder (image encoder + text encoder on clinical captions). The index is built from all 911 training images; validation images are excluded from the index to prevent retrieval-level leakage. $\alpha$ is swept over $\{0.5, 0.7, 0.8, 0.9, 1.0\}$ after encoder selection on the locked validation set ($\alpha$ selection introduces known hyperparameter bias, reported in Section IV-B). The retrieval score is:

$$\text{score}(q,i) = \alpha\,\cos\!\big(\mathbf{e}_q^{img}, \mathbf{e}_i^{img}\big) + (1-\alpha)\,\cos\!\big(\mathbf{e}_q^{txt}, \mathbf{e}_i^{txt}\big), \quad (1)$$

where $\alpha$ controls image-to-text weight ($\alpha$=1: image only; $\alpha$=0: text only). Text embeddings $\mathbf{e}^{txt}$ are pre-computed from SkinCAP captions for index images only; query images at inference have no ground-truth captions, so the text term is omitted ($\mathbf{e}_q^{txt}$=$\mathbf{0}$) and the score reduces to the image component scaled by $\alpha$. We evaluate five encoder configurations (Table II).

**Training sample format.** Each Stage 2 sample uses $K$=1 retrieved reference during training ($K$=3 exhausts the 15.92 GB VRAM budget under gradient computation). All RAG inference evaluations use $K$=3 references in the prompt (VRAM is sufficient without gradient computation). The prompt prepends `[ref_img] Reference N: <disease>` for each reference before the query, and the model answers with the query's disease label. Reference embeddings are pre-computed before loading Qwen3-VL.

**Output parsing.** Raw outputs are parsed by stripping `<think>` blocks, anchor extraction (`Diagnosis:`), and fuzzy string matching (thefuzz library, $\geq$91% similarity) against the 10 disease names; unmatched outputs count as incorrect.

### C. Merged-Init for Task Switching

**Way 1 (Checkpoint Init)** reuses Stage 2 LoRA adapters for caption training; caption gradients overwrite disease representations, causing catastrophic interference. **Way 2 (Merged-Init)** first merges Stage 2 ($W' = W + (s/r)BA$, where $s$ is the LoRA scaling factor, distinct from the retrieval weight $\alpha$ in Eq. 1; this freezes disease knowledge in full-rank weights) then attaches fresh adapters for caption training, eliminating gradient coupling between tasks. Since the merge is algebraically exact, $W'$ behaves identically to the fully fine-tuned model—no task-A capability is sacrificed—and the principle applies to any adapter parameterisation whose weights can be folded losslessly into the base model (LoRA, adapter layers, etc.). Way 2 achieves BLEU-4 = 29.33 vs. 9.82 (Table VI).

## IV. Experiments

### A. Dataset and Setup

**SkinCAP** [8] contains 4,000 dermatological images with disease labels and clinical captions. We filter to 10 disease classes across four groups (Inflammatory/Autoimmune, Benign, Malignant, and Acne Vulgaris; 1,010 images) with a stratified split: 911 train / 99 validation (locked; never

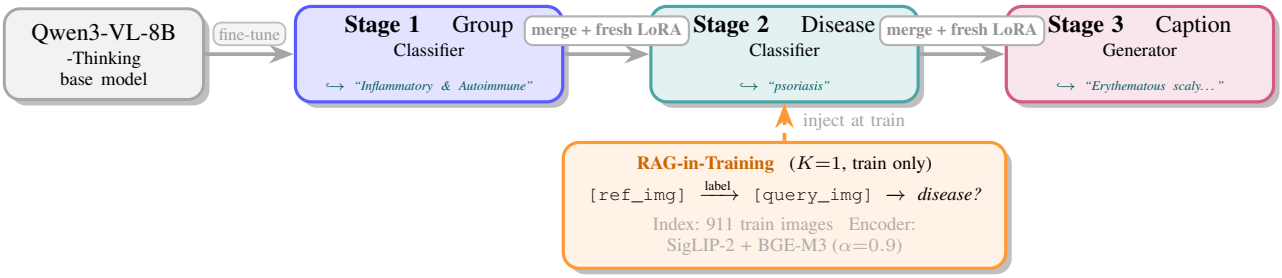



Fig. 1. HIKARI three-stage pipeline. **Merged-Init** (white badges on arrows) absorbs LoRA adapters into base weights before each stage transition, preventing catastrophic interference. **RAG-in-Training** (orange, train only) injects one retrieved reference image per Stage 2 sample. Teal text shows example outputs at each stage.

regenerated). Images are resized to 672×672 px using Lanczos filtering.

The 10 classes span clinical urgency: benign (Melanocytic Nevi, Acne Vulgaris), pre-malignant (SCCIS/Bowen's Disease), malignant (BCC), and inflammatory/autoimmune conditions requiring systemic management (Psoriasis, Lupus Erythematosus, Lichen Planus, Scleroderma, Photodermatoses, Sarcoidosis). This heterogeneity—benign vs. malignant vs. systemic—makes the 10-class task clinically representative and computationally challenging.

Validation counts range from 7 (Sarcoidosis) to 13 (Psoriasis, BCC); Sarcoidosis and Acne Vulgaris are square-root oversampled during training to reduce class-frequency bias.

**Hyperparameter selection.** LoRA rank ($r$=16), learning rate ($2\times10^{-4}$), and optimizer (AdamW-8bit) follow standard VLM LoRA fine-tuning practice [3] and were *not* optimised on the validation set. Batch sizes reflect VRAM constraints (batch 8 for Stage 2 without gradient accumulation; batch 2 with 4-step accumulation elsewhere). $K$=1 at training is the maximum feasible under gradient computation on 15.92 GB VRAM; $K$=3 at inference fits without gradient overhead. *Only* $\alpha$ (swept over $\{0.5, 0.7, 0.8, 0.9, 1.0\}$) was tuned on the locked validation set after encoder selection, introducing the selection bias acknowledged in Section IV-B.

### B. RAG Encoder Comparison

Table II compares five encoder combinations (Cascaded FT, $K$=3). R2 (SigLIP-2 [21]+BGE-M3 [6], $\alpha$=0.9) achieves the best accuracy (Wilson 95% CI [71–87%]) and is selected for RAG-in-Training. Image-dominant weighting outperforms balanced weighting because clinical skin lesions are visually discriminated; equal-weight text captions introduce retrieval noise. R3 (Jina-CLIP-v2+MedCPT, $\alpha$=0.7) comes within 1.01 pp of R2 (78.79%, CI [70–86%]), suggesting domain-specific text embeddings partially compensate for weaker image-only similarity; R0 and R4 both reach 72.73% (CI [63–81%]), R1 71.72% (CI [62–80%]). $\alpha$ was tuned after encoder selection on the locked validation set ($\alpha$ selection is a known bias source; Best-column values are selection-biased upper bounds—a held-out set would yield unbiased estimates).

TABLE II
RAG ENCODER COMPARISON – CASCADED FT (99 VAL, $K$=3).

| ID | Encoder (Image + Text) | $\alpha$=0.5 | Best |
|---|---|---|---|
| R0 | CLIP ViT-B/32 [5] + None | 72.73% | 72.73% |
| R1 | CLIP + ClinicalBERT [18] | 69.70% | 71.72% ($\alpha$=0.9) |
| **R2** | **SigLIP-2 [21] + BGE-M3 [6]** | **74.75%** | **79.80%** ($\alpha$=0.9) |
| R3 | Jina-CLIP-v2 [19] + MedCPT [7] | 66.67% | 78.79% ($\alpha$=0.7) |
| R4 | Nomic Vision + Nomic Text [20] | 67.68% | 72.73% ($\alpha$=0.9) |

### C. Main Classification Results

Table III reports classification accuracy on the locked 99-sample validation set. CNN baselines peak at 65.00% (YOLOv11; CI [55.2–73.7%]); ResNet-50 collapses to 8.00% (CI [4.1–15.0%]) on this long-tail distribution. Seven frontier VLMs score 12.87–50.51% (LLaMA/Gemini: raw image only; Qwen base: R0 context)—none reaches adequate performance without RAG-aware training.

The CNN spread (ResNet-50: 8.00%→YOLOv11: 65.00%) reflects dataset difficulty: ResNet-50 collapses under the 10-class long-tail distribution, while YOLOv11's multi-scale feature pyramid provides better visual grounding. Even at 65.00%, the best CNN falls 20.86 pp below HIKARI, indicating that fine-grained inflammatory lesion classification benefits from language-conditioned visual reasoning unavailable to CNNs. Among zero-shot VLMs, Qwen2.5-VL-7B (50.51%) outperforms the larger Qwen3-VL-8B (33.33%) under identical R0 retrieval context, suggesting generation-level differences in dermatology exposure during pre-training; both remain far below any fine-tuned variant.

**HIKARI with RAG-in-Training** achieves 85.86% (95% Wilson CI [77.7–91.4%]), surpassing the strongest CNN baseline (YOLOv11, 65.00%) by 20.86 pp, inference-only RAG (Cascaded FT, R2 $\alpha$=0.9: 79.80%, CI [70.8–86.5%]) by 6.06 pp, and the single-image FT baseline (74.00%; CI [64.3–81.4%]) by 11.86 pp. The 6.06 pp advantage is directional but not statistically significant (95% CI: $-4.4$ to $+16.5$ pp on $n$=99); larger held-out sets are needed. Cascaded FT with R0 context scores only 59.38%—lower than Single-Image FT without RAG (74.00%)—confirming that reference context only helps when the model is trained to use it.

TABLE III
MAIN CLASSIFICATION RESULTS (99 VAL SAMPLES, ALL AUTHOR-EVALUATED). CNN BASELINES TRAINED ON SAME DATASET (NO RAG). LLAMA/GEMINI: RAW IMAGE ONLY; QWEN ZERO-SHOT: CLIP (R0) CONTEXT, $K$=3; NO FINE-TUNING. ALL RAG INFERENCE USES $K$=3; RAG-IN-TRAINING TRAINS WITH $K$=1. ENCODER IDS (R0, R2) DEFINED IN TABLE II. ⋆GEMINI-3.1-PRO-PREVIEW ACCESSED VIA API, APRIL 2025. LLAMA-4 SCOUT AND MAVERICK-17B BOTH SCORED 12.87%; COMBINED AS ONE ROW.

| Model | RAG Train | RAG Infer. | Acc. |
|---|---|---|---|
| *CNN Baselines (fine-tuned, no RAG)* | | | |
| ResNet-50 [14] | – | – | 8.00% |
| YOLOv8 [16] | – | – | 39.00% |
| EfficientNet-B0 [15] | – | – | 64.00% |
| YOLOv11 [17] | – | – | 65.00% |
| *Zero-Shot Frontier Models (no fine-tuning)* | | | |
| LLaMA-4 (Scout/Maverick-17B) [22] | – | – | 12.87% |
| Gemini-2.5-Flash [23] | – | – | 25.74% |
| Gemini-3.1-Pro-preview⋆ [23] | – | – | 23.23% |
| Gemini-2.5-Pro [23] | – | – | 30.69% |
| Qwen3-VL-8B [11] | – | R0 | 33.33% |
| Qwen2.5-VL-7B [9] | – | R0 | 50.51% |
| *Domain Fine-Tuned (No Training RAG)* | | | |
| Single-Image FT | – | – | 74.00% |
| *Inference-Only RAG (Cascaded FT, no RAG training)* | | | |
| Cascaded FT (R0) | – | R0 | 59.38% |
| Cascaded FT (R2, $\alpha$=0.5) | – | R2 | 74.75% |
| Cascaded FT (R2, $\alpha$=0.9) | – | R2 | 79.80% |
| *RAG-in-Training (Ours)* | | | |
| RAG-in-Training (R2 train, R2) | R2 K=1 | R2 | 82.83% |
| **RAG-in-Training (R2 train, R0)** | **R2 K=1** | **R0** | **85.86%** |

### D. Per-Disease Analysis

Table IV reports per-disease sensitivity. Classes with $n$<10 (*) are statistically fragile (each error ≥10 pp).

TABLE IV
PER-DISEASE SENSITIVITY [%] WITH WILSON 95% CIS. *$n$<10: EACH ERROR ≥10 PP. CASFT = CASCADED FT (R2, $\alpha$=0.9).

| Disease | $n$ | CasFT [%] | HIKARI [%] | Δ |
|---|---|---|---|---|
| Acne Vulgaris | 8* | 100 [68–100] | 100 [68–100] | 0 |
| BCC | 13 | 76.9 [50–92] | 100 [77–100] | +23 |
| Lichen Planus | 9* | 88.9 [56–98] | 88.9 [56–98] | 0 |
| Lupus Erythema. | 9* | 77.8 [45–94] | 55.6 [27–81] | −22 |
| Melanocytic Nevi | 12 | 91.7 [65–99] | 100 [76–100] | +8 |
| Photodermatoses | 8* | 62.5 [31–86] | 75.0 [41–93] | +13 |
| Psoriasis | 13 | 61.5 [36–82] | 100 [77–100] | +38.5 |
| Sarcoidosis | 7* | 57.1 [25–84] | 14.3 [3–51] | −43 |
| Scleroderma | 8* | 75.0 [41–93] | 87.5 [53–98] | +13 |
| SCCIS | 12 | 100 [76–100] | 100 [76–100] | 0 |
| **Overall** | 99 | **79.8 [71–87]** | **85.9 [77–91]** | **+6.1** |

RAG-in-Training strictly improves 5 of 10 diseases; five diseases reach 100% under HIKARI, led by psoriasis (+38.5 pp). Sarcoidosis drops sharply (57.1%→14.3%, CI [3–51%] on $n$=7; only 1 of 7 samples correctly classified under HIKARI).

**Sarcoidosis failure mode.** Sarcoidosis exhibits bimodal visual presentation (erythematous plaques vs. papular), causing high within-class diversity (mean pairwise Hamming distance 2.56 vs. 1.91 for nevi). References spanning the wrong subtype actively misdirect prediction (*reference-label bias*). Critically, square-root oversampling mitigates class-frequency bias but not within-class appearance diversity: Acne Vulgaris (oversampled, $n$=8) reaches 100%, while Sarcoidosis (oversampled, $n$=7) degrades to 14.3%—confirming the failure is retrieval-induced, not frequency-induced. Lupus (−22 pp, $n$=9) shares this risk. A practical vulnerability indicator is high mean within-class visual dissimilarity: classes whose retrieved references span visually distinct subtypes are at elevated risk regardless of training strategy. Contrastive hard-negative training or majority voting across separate single-reference inferences are the primary mitigations.

**High-gain diseases.** Psoriasis achieves the largest gain (+38.5 pp; 8/13→13/13): its highly distinctive erythematous scaly plaques are strongly reinforced by retrieved references, enabling confident classification even in atypical presentations. BCC gains +23 pp (10/13→13/13): retrieved nodular images highlight the raised pearly border that distinguishes BCC from melanocytic nevi, aligning with standard ABCDE dermoscopy criteria. Three diseases show zero delta (Acne Vulgaris, Lichen Planus, SCCIS)—each was at or near ceiling under Cascaded FT, indicating RAG-in-Training saturates quickly for visually homogeneous classes that retrieved references consistently identify correctly even without RAG-aware training.

### E. Merged-Init Ablation (Stages 2 & 3)

Table VI validates Merged-Init: Stage 2 gains +4.6 pp (85.9% vs. 81.3% base init); Stage 3 achieves BLEU-4 = 29.33 (3× Checkpoint Way 1), ROUGE-1 = 53.6 (+14.7 pp), and BERTScore = 92.1, confirming that absorbing prior-stage adapters into full-rank weights prevents catastrophic interference. The Checkpoint failure is qualitatively severe: Stage 3 Checkpoint outputs exhibit label-hallucination artifacts (disease labels interspersed within caption text), producing near-zero lexical overlap with reference captions and rendering BERTScore $F_1$ meaningless (reported as N/A; see Table caption). Merged-Init eliminates this entirely: all generated outputs are well-formed clinical captions. The BLEU-4 improvement (3×) and +14.7 pp ROUGE-1 therefore reflect a qualitative shift from degenerate generation to coherent medical text, not merely metric fluctuation. Table V illustrates this contrast with a representative example.

### F. Attention Map Analysis

Attention is averaged over the last four transformer layers (last-input token to all `<|image_pad|>` positions), producing a patch saliency map bilinearly upsampled to full resolution.

Fig. 2 shows the base model distributes attention diffusely across the full image (skin, hair, background), while HIKARI concentrates on clinically discriminating regions—the pigmented lesion border for nevi; the annotated nodular lesion for BCC. This attention shift aligns with clinical differential diagnosis: BCC is distinguished from nevi primarily by raised

TABLE V

REPRESENTATIVE STAGE 3 OUTPUTS FOR THE SAME QUERY IMAGE (PSORIASIS). CHECKPOINT OUTPUT EXHIBITS LABEL-HALLUCINATION ARTIFACTS; MERGED-INIT PRODUCES COHERENT CLINICAL TEXT. OUTPUTS ARE REPRESENTATIVE OF THE OBSERVED FAILURE PATTERN.

| **Init** | **Generated Output (excerpt)** |
|---|---|
| Checkpoint | *"Psoriasis. Lichen Planus. BCC psoriasis. Lichen planus acne psoriasis lichen psoriasis BCC psoriasis lichen planus…"* |
| **Merged-Init** | *"Erythematous, well-demarcated scaly plaques with silvery scaling distributed over the extensor surfaces, consistent with plaque-type psoriasis."* |

TABLE VI

MERGED-INIT ABLATION FOR STAGE 2 (CLASSIFICATION ACCURACY, $n$=99) AND STAGE 3 (CAPTION METRICS, $n$=99). BOTH STAGE 2 ROWS USE RAG-IN-TRAINING ($K$=1 TRAIN, $K$=3 INFER); THE COMPARISON ISOLATES THE STAGE 1 MERGED-INIT EFFECT. $\mathrm{BS}_{F1}$: BERTSCORE $F_1$ [24] (MEDCPT ENCODER). †CHECKPOINT OUTPUTS EXHIBITED SEVERE LABEL-HALLUCINATION ARTIFACTS; $\mathrm{BS}_{F1}$ IS NOT REPORTED AS IT WOULD NOT REFLECT CAPTION QUALITY.

| **Stage** | **Init** | **Acc.** | **B-4** | **R-1** | $\mathbf{BS}_{F1}$ |
|---|---|---|---|---|---|
| Stage 2‡ | Fresh Init (base) | 81.3% | – | – | – |
| | **Merged-Init** | **85.9%** | – | – | – |
| Stage 3 | Checkpoint | – | 9.82 | 38.9 | N/A† |
| | **Merged-Init** | – | **29.33** | **53.6** | **92.1** |

B-4: BLEU-4; R-1: ROUGE-1; BS: BERTScore $F_1$

‡Stage 2 accuracy uses R0 (CLIP ViT-B/32) retrieval at inference.

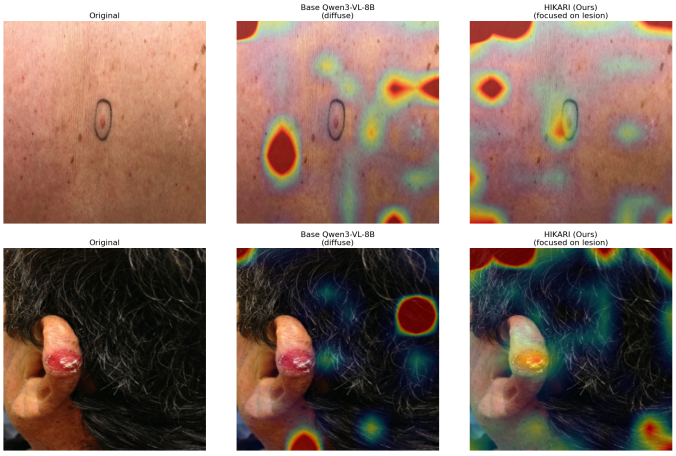



Fig. 2. Attention maps for melanocytic nevi (top) and BCC (bottom). Each row: **Original**; **Base** Qwen3-VL-8B (diffuse); **HIKARI** (concentrated on the pigmented lesion for nevi; on the annotated nodule for BCC). Warmer colours = higher attention.

pearly border morphology, which HIKARI's localised attention correctly targets; for melanocytic nevi, the model focuses on the central pigmented lesion, consistent with ABCDE dermoscopy criteria (asymmetry, border, colour, diameter, evolution). The base model's diffuse background attention suggests it relies on spurious contextual features rather than lesion-specific morphology.

**Proxy-mask IoU.** Using hand-drawn annotations as noisy proxies ($n$=10): HIKARI IoU = 0.175 vs. base = 0.167 (Dice: 0.279 vs. 0.268). These values are low and proxy-noise dominated; they support only the qualitative claim that HIKARI attention is more lesion-directed. Pixel-level expert annotations are required for any quantitative localisation claim.

### G. Inference Speed

Table VII benchmarks three inference engines on RTX 5070 Ti using the same merged HIKARI weights. LMDeploy and TensorRT-LLM were excluded due to VRAM and driver constraints.

TABLE VII

INFERENCE SPEED (MEAN MS/IMAGE ± STD) ON RTX 5070 TI, MEASURED ON THE 99-SAMPLE VALIDATION SET. SPEEDUP VS. UNSLOTH-4BIT BATCH = 1 (TRAINING BASELINE).

| | **Stage 2 (Classif.)** | | **Stage 3 (Caption)** | |
|---|---|---|---|---|
| **Engine** | bs = 1 | bs = 4 | bs = 1 | bs = 4 |
| Unsloth-4bit | 1096 ±39 | 500 ±10 | 6699 ±156 | 3003 ±80 |
| vLLM-BnB4 [12] | 480 ±44 | 179 ±24 | 2957 ±9 | 1094 ±3 |
| **SGLang-FP8** [13] | **331 ±3** | **110 ±2** | **1695 ±4** | **584 ±1** |

Speedup (bs = 4 vs. Unsloth bs = 1): vLLM 6.1×; SGLang 10.0× (S2), 11.5× (S3). SGLang: FP8 online quant.; vLLM: BitsAndBytes 4-bit.

SGLang-FP8 (batch = 4) achieves 110 ms/image for classification (10×) and 584 ms/image for captions (11.5×) over the Unsloth training baseline. vLLM-BnB4 offers 6.1× at 4-bit fidelity.

## V. DISCUSSION

RAG-in-Training confirms that reference context only helps when the model is *trained to use it* (CI $-4.4$ to $+16.5$ pp on $n$=99; gains are directional). Merged-Init's 3× BLEU-4 gain and +14.7 pp ROUGE-1 apply to any multi-task LoRA pipeline where task objectives conflict. SGLang-FP8 achieves 110 ms/image (10× over training baseline; Table VII), making HIKARI practical on a single RTX 5070 Ti. *Reference-label bias* (Sarcoidosis $-43$ pp, Lupus $-22$ pp) is a systemic risk for any retrieval-augmented diagnostic system with high intra-class appearance diversity, not a limitation specific to HIKARI; it motivates contrastive retrieval strategies as a general design principle.

**Encoder flexibility.** HIKARI trained on R2 (SigLIP-2+BGE-M3) performs best with R0 (CLIP ViT-B/32, 85.86% vs. 82.83%; 3.03 pp gap, CI: $-7.1$ to $+13.1$ pp, includes 0), suggesting RAG-in-Training builds an encoder-agnostic context-reading strategy. CLIP's simpler embeddings may retrieve semantically cleaner references for this closed-world task. Deployment can use a lightweight encoder without retraining; a larger held-out set is required to confirm.

**Caption clinical utility.** BERTScore $F_1$ [24] computed with a MedCPT encoder [7] provides a domain-specific semantic similarity proxy beyond surface lexical overlap (BLEU/ROUGE); higher scores reflect alignment with clinical

terminology in reference captions. Clinician evaluation of caption acceptability and actionability for clinical decision support is out of scope for this prototype and is a primary future-work target.

**Deployment pathway.** HIKARI's modular design supports incremental clinical integration: Stage 2 can be deployed first for triage classification; Stage 3 is added only when clinical narrative generation is required, with no retraining of the classifier. The encoder-agnostic inference property (R0 matching R2 at $n$=99) means the RAG index can be rebuilt with any available encoder without retraining HIKARI's weights—a significant advantage in resource-constrained clinical settings where CLIP ViT-B/32 may be the only available encoder. SGLang-FP8 at 110 ms/image (Stage 2, batch = 4) supports real-time triage throughput on a single RTX 5070 Ti, making single-GPU deployment feasible outside GPU-cluster environments.

**Future work.** Four directions follow directly from current limitations: (1) *Contrastive hard-negative retrieval*—retrieve visually similar but class-different negatives during training to suppress reference-label bias in high within-class diversity diseases (Sarcoidosis, Lupus Erythematosus); (2) $K$=1 *inference ablation*—evaluate the trained model with a single reference at inference to disentangle distribution-gap reduction from reference-count effects, isolating the RAG-in-Training contribution from the $K$:1→3 gap; (3) *Cross-domain validation*—apply HIKARI to radiology, ophthalmology, and oral mucosal disease by rebuilding the RAG index alone, without modifying training, to empirically confirm the domain-agnostic architecture claim; (4) *Clinician caption evaluation*—domain expert assessment of caption acceptability, clinical accuracy, and actionability for decision support pipelines, using BERTScore $F_1$ as a pre-screening filter to identify candidates for expert review.

**Limitations.** Statistical power is low ($n$=99; 7–13 per class); the 6.06 pp gain is directional only and per-disease results for $n$<10 are fragile. A train-inference gap exists: the model trains with $K$=1 reference but evaluates with $K$=3; a dedicated $K$=1 inference ablation is needed to isolate the reference-count contribution from the distribution-gap reduction. Proxy-mask IoU (0.175 vs. 0.167, $n$=10 annotated images) is noise-dominated; pixel-level expert annotations are required for any quantitative localisation claim. Cross-domain validation (radiology, ophthalmology, oral mucosal disease) and larger held-out sets (≥200 per class) are the primary next steps.

## VI. Conclusion

HIKARI achieves 85.86% (CI [77.7–91.4%]) on a 10-class SkinCAP validation set via **RAG-in-Training**, outperforming the strongest CNN baseline (YOLOv11, 65.00%) by 20.86 pp, single-image FT by 11.86 pp, and inference-only RAG by 6.06 pp (directional). Seven frontier VLMs without fine-tuning score 12.87–50.51%, confirming that RAG-aware training is essential. **Merged-Init** eliminates inter-task gradient interference, yielding 3× BLEU-4 (9.82→29.33) and ROUGE-1: 38.9→53.6 (+14.7 pp). SGLang-FP8 [13] delivers 110 ms/image (10× speedup on RTX 5070 Ti). Cross-domain validation and clinician caption evaluation are the primary next steps.


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